

・いろいろな数列の和

(例11) 次の和 S を求めよ

$$S = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots + \frac{1}{n(n+1)}$$

point

分数 a の和 $\rightarrow$ 差の形をつくる $\circ \circ \circ$ つくれたらいいな

$$\text{第 } k \text{ 項 } \frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$$

$$\begin{aligned} S &= \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1} \\ &= \frac{n}{n+1} \end{aligned}$$

(例12) 次の和 S を求めよ

$$S = \frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \cdots + \frac{1}{(2n-1)(2n+1)}$$

point

差の形をつくれたらいいな

$$\text{第 } k \text{ 項 } \frac{1}{(2k-1)(2k+1)} = \frac{1}{2} \left(\frac{1}{2k-1} - \frac{1}{2k+1} \right)$$

本当に、成り立つのかな?

$$\frac{1}{(2k-1)(2k+1)} = \frac{1}{2} \cdot \frac{(2k+1) - (2k-1)}{(2k-1)(2k+1)}$$

つじつまあわせ

$$\begin{aligned} S &= \frac{1}{2} \left(\frac{1}{1} - \frac{1}{3} \right) + \frac{1}{2} \left(\frac{1}{3} - \frac{1}{5} \right) + \frac{1}{2} \left(\frac{1}{5} - \frac{1}{7} \right) + \cdots + \frac{1}{2} \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right) \\ &= \frac{1}{2} \left(\frac{1}{1} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{7} \right) + \cdots + \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right) \\ &= \frac{1}{2} \left(1 - \frac{1}{2n+1} \right) \\ &= \frac{n}{2n+1} \end{aligned}$$

(例13) 次の和 S を求めよ

$$S = \frac{1}{1 + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \cdots + \frac{1}{\sqrt{n} + \sqrt{n+1}}$$

point

 $\frac{1}{\sqrt{+}\sqrt{+}}$ の形 $\rightarrow$ 有理化して差の形をつくる

$$\text{第 } k \text{ 項 } \frac{1}{\sqrt{k} + \sqrt{k+1}} = \frac{\sqrt{k} - \sqrt{k+1}}{k - (k+1)} = \sqrt{k+1} - \sqrt{k}$$

$$\begin{aligned} S &= (\sqrt{2} - 1) + (\sqrt{3} - \sqrt{2}) + (\sqrt{4} - \sqrt{3}) + \cdots + (\sqrt{n+1} - \sqrt{n}) \\ &= \sqrt{n+1} - 1 \end{aligned}$$

(例14) 次の和 S を求めよ

$$S = 1 \cdot 1 + 2 \cdot 2 + 3 \cdot 2^2 + \cdots + n \cdot 2^{n-1}$$

point

(等差) $\times$ (等比) 型の和の求め方は

等比数列の和を求める方法と同じ

$$S = 1 \cdot 1 + 2 \cdot 2 + 3 \cdot 2^2 + \cdots + n \cdot 2^{n-1}$$

$$-12S = \begin{array}{ccccccc} \times 2 & \times 2 & \times 2 & \times 2 & \times 2 & & \\ 1 \cdot 2 & 2 \cdot 2^2 & 3 \cdot 2^3 & \cdots & (n-1) \cdot 2^{n-1} & + & n \cdot 2^n \end{array}$$

$$-S = 1 + 2 + 2^2 + \cdots + 2^{n-1} - n \cdot 2^n$$

よて

$$-S = \frac{2^n - 1}{1 - 2} - n \cdot 2^n$$

$$\therefore S = (n-1)2^n + 1$$

$$\begin{aligned} T &= 1 + 2 + \cdots + 2^{n-1} \\ -12T &= 2 + \cdots + 2^{n-1} + 2^n \\ -T &= 1 - 2^n \\ \therefore T &= 2^n - 1 \end{aligned}$$